The Kernel That Builds Its Own Laser: Five Theorems on Covariant Modified Inertia, from Specification to Closure

Carl P. Zimmerman · Briar Creek Tech · 2026-07-03

Abstract

Whether MOND can be realized as a genuine modification of *inertia* — covariant, ghost-free, and consistent with Solar-System and binary data — has been open since Milgrom posed it in 1994. This note completes a five-theorem chain that takes the question from prohibition to specification to constructive closure, each step adversarially verified with committed, runnable code. **Theorems I–IV** (published separately) close the local and passive routes: local Hermitian higher-derivative actions carry the Ostrogradsky ghost; field-theoretic completions are fixed by Cassini; nonlocal kernels over *passive* baths carry the anti-MOND sign — a premise shown to be a *state* clause, with population inversion the unique sign-viable escape and free fields and the de Sitter horizon provably unable to supply it; and local PT-quantized preferred-frame theories produce a frequency law, never $\mu(a/a_0)$, with healthy-phase softening capped at 50% by an exceptional point. These four theorems *force* the shape of any surviving realization: a causal, nonlocal-in-time kernel fed by an inverted (non-KMS) anharmonic medium, with net inverted spectral weight strictly above the galactic dynamical band. **We then built it.** The construction — $m_{eff} = m[1 - S(|a|/a_0)R(\omega)]$, one inverted line at $\nu_2 \in [1.1\times10^6, 9.9\times10^7]\ H_0$, saturation carrying the framework interpolation, coupling mass-proportional and tuned to cancel inertia to ≤1% — collapses exactly (verified to 10^{-12}) to $g_{obs} = \sqrt{g_{bar}^2 + g_{bar} \cdot a_0}$ on circular orbits, and *passes* every gate that killed its predecessors: it is causal (Kramers–Krönig with sign-indefinite spectral weight; poles in the lower half-plane), stable (the deep-MOND limit $|\delta m| \to m$ is precisely the marginal-stability edge, and the saturation bound $S \leq 1$ is precisely the stability clamp), Solar-System- and pulsar-safe by five to twenty-nine orders, energetically affordable (pump demand 25–470× *below* the available Λ-scale flux — the anticipated energetics kill does not exist), and universal to 10^{-8} dex. **Theorem V** then closes it. By the same Kramers–Krönig lock that lets the line cancel inertia in-band, the inverted line is a **super-threshold amplifier at its own frequency**, with irreducible per-radian gain $\geq 0.94 \cdot S$ surviving maximal inhomogeneous dilution — and the cold interstellar medium's acoustic continuum (wavelengths 0.001–0.09 pc, comfortably fluid) spans *every* admissible line placement. In steady state the dichotomy is exhaustive: either the inversion **clamps** at threshold, capping the delivered deep-MOND boost at ~1.4 versus the observed ~10 — erasing the MOND limit *in exactly the cold-HI volumes where the radial-acceleration relation is measured*, while gas-poor spheroidals keep full a_0, destroying the tuning, the universality, and the construction's own 0.108-dex fit; or the pump refresh outruns the stimulated drain at 10^5–10^7 times the priced budget, and the energetics gate fails jointly with this one (margin ≥ 350). A 25-point scan of the surviving (ν_2, σ) box finds **no live subregion**. The theorem's shape mirrors its precursors: "in-band" is wherever the weight sits, and in a galaxy something always oscillates there — for a continuum medium, everything does. The medium the four theorems demand builds its own laser, and the galaxy is the cavity. Scope is stated exactly: this closes the inverted-line modified-inertia realization — not "modified inertia" *simpliciter* (one passive-anharmonic corner remains

on the prior sign-theorem adjudication), and not the a_0 reframing or its live empirical tests, which are untouched. Two candidate observables raised en route (a wide-binary $\gamma \rightarrow 1.28$ and a binary separation-gap sweep) are **withdrawn** with the kernel — the second nulled at ×2900 by spectral hole-burning during verification, which also deflated the headline margins ×2.5 (the kill survives at ×3.4 in the most hostile corner under tenfold damping stress). Every number in this chain is reproducible from committed scripts; every correction found in verification is reported in the text that states the result.

1. The chain to date

The five results, in logical order (I–IV published; detailed statements and scripts in the cited records):

- **I (Ostrogradsky).** Local Hermitian $L(x, \dot{x}, \ddot{x})$ with genuine acceleration dependence has a nondegenerate Hessian and the classical ghost; degenerate cases reduce exactly to $L(x, \dot{x})$, whose inertia can never depend on acceleration (the variational half of this statement is the G0 theorem, committed).
- **II (Cassini).** Field-theoretic completions carry a static slip fixed by $|\gamma-1| \leq 2.3\times10^{-5}$ — a wall that now extends across the full Blanchet–Skordis lineage.
- **III (the state clause).** Nonlocal kernels over passive baths give $\delta m \geq 0$ — anti-MOND — at any temperature; the premise is the bath's *state*, not ghost-freedom; population inversion flips the sign; free fields are state-blind and the Gibbons–Hawking horizon is a thermostat (zero extractable work), so the inverted medium is necessarily a *new posit* (10.5281/zenodo.21139029).
- **IV (the frequency law).** Local preferred-frame PT-quantized theories yield $\mu\_eff = 1 - (\Omega/w\_eff)^2$ — selected by frequency, never by acceleration — with unbroken-PT softening capped at 50% and deep-MOND orbits failing by non-existence at an exceptional point (10.5281/zenodo.21148494).

Together these do something unusual for no-go results: they *specify* the survivor. Any realization must be a causal retarded kernel; its spectral density must carry net **inverted** weight; that weight must sit **strictly above** the galactic band (in-band weight is capped at $|Im/Re| < 10^{-4}$ by orbit-decay bounds; below-band weight has the wrong shape by 3.7 dex); its integrated weight must cancel inertia to ≤1% (else a_0 drifts by 0.125 dex); and its saturation must run on $|a|$ — the accelerometer variable that the de Sitter–Unruh temperature supplies naturally. Five forced properties. One object.

2. The construction, and what it survived

We built the minimal realization: a single inverted line at ν_2 with homogeneous width $\Gamma = 3H_0$ (the phase-pinning floor), Gaussian inhomogeneous profile (the Lorentzian variant dies on its tails by 6×10^6), mass-proportional coupling tuned to the −m cancellation, and saturation $S(|a|/a_0) = 1 - \mu\_fw$ inserted to carry the framework interpolation (per the standing decoy-family lesson, the shape is a posit and is flagged as one). On circular orbits it collapses to $g\_obs = \sqrt{(g\_bar^2 + g\_bar \cdot a_0)}$ exactly — the 0.108-dex SPARC law by construction, satisfying Milgrom's 2022 circular-orbit constraint.

The gates it passed are the graveyard of its predecessors. *Causality:* sign-indefinite spectral weight is fully causal (analyticity constrains poles, not signs; gain media are laboratory physics), Kramers–Krönig verified numerically to 10^{-3}. *Stability:* dressed-propagator poles remain in the lower half-plane for $|\delta m| < m$ — and the deep-MOND limit approaches that edge exactly as the saturation bound $S \le 1$ clamps it: the framework's interpolation turns out to be the stability boundary written as phenomenology. *Solar System:* Saturn at 7.6×10^{-14}, LLR at 10^{-20}, pulsars at 10^{-29}; the line's rolloff even suppresses the Cassini-quadrupole inheritance that burdens the modified-gravity realizations. *Energetics* — the gate we expected to be fatal: the pump power to maintain the inversion against spontaneous refresh is 25–470× *below* the Λ-scale flux $\rho\_\Lambda c^2 H_0$, computed independently twice. *Universality:* band-flatness to 1.8×10^{-8} dex. For one working day, the object all four theorems pointed at existed on paper and would not die.

3. Theorem V: the gas-clamp catastrophe

The same tuning that lets the line cancel inertia is a Kramers–Krönig lock on its dissipative part: the integrated weight is fixed, so the at-line gain is irreducible — after *maximal* admissible inhomogeneous dilution (profile positivity caps $\sigma \le (\nu_2 - W)/3$), any oscillator with eigenfrequency inside the line is anti-damped at a per-radian rate $\ge 0.94 \cdot S$, per worldline, with no phase-matching or collectivity required (gravitationally-mediated cooperative channels are independently dead at 6×10^{-8} — the kill needs no Dicke physics, and inhomogeneous dephasing, which does quench cooperativity, cannot touch single-mode gain).

What oscillates there? At $\nu_2 \in [1.1\times10^{6}, 9.9\times10^{7}]\ H_0$ — periods of a millennium to 80 millennia — the cold neutral interstellar medium's acoustic and magnetosonic modes form a **continuum** across the entire box (wavelengths 0.001–0.09 pc; fluid regime by mean-free-path ratios of 91–8200), with realistic losses ≤ 0.2 per radian on exactly the worldlines where the saturation is deep ($S \ge 0.7$). Gain over loss: ×1295 at the box floor, ×14 at its ceiling — ×130 to ×3.4 under a further tenfold damping stress. Super-threshold everywhere.

Steady state then admits exactly two branches, both computed. **Clamp:** the stimulated drain pins the inversion at threshold, and the *delivered* spectral weight collapses to $r \le 0.29$ even in the most hostile corner — capping the deep-MOND boost at $1/(1-r) \approx 1.4$ against the observed $g\_obs/g\_bar \sim 10$. The MOND limit is erased *qualitatively*, and erased selectively: in the HI- and CNM-bearing disk volumes where the radial-acceleration relation is actually measured, while gas-poor dwarf spheroidals — no acoustic continuum — retain full a_0. The ≤1% tuning, the 0.079-dex universality budget, and the construction's own SPARC fit are all destroyed at once. **No-clamp:** holding the tuning requires refresh at $\sim 0.45 \cdot \nu_2$, exceeding the priced pump budget by 1.5×10^{5}–1.5×10^{7} — the energetics gate, which the construction had passed, fails jointly by a factor ≥ 350. A 25-point logarithmic scan of the (ν_2, σ) box: zero subregions alive on either branch. The box is empty.

Theorem V (gas-clamp form). *An inverted spectral line whose integrated weight is tuned to deliver the in-band inertia deficit $\Delta = -mS$ is, by Kramers–Krönig, a super-threshold amplifier at its own frequency; the cold-ISM acoustic continuum spans every admissible placement of that line; and in steady state either the inversion clamps — erasing the deep-MOND limit precisely where it is measured — or the refresh required to prevent clamping exceeds the available pump budget by more than two orders. No placement escapes both.* Assumptions: per-worldline O(1) mass-proportional coupling (forced by the tuning theorem); $\Gamma \sim 3H_0$ (forced by the pinning floor); a cold neutral medium with n ≥ 10 cm^{-3}, $T \leq 100$ K, losses ≤ 2/radian (tenfold stress included); and the framework interpolation's own Newtonian tail (unchangeable without abandoning the interpolation).

4. Verification, corrections, and withdrawn observables

The adversarial pass on Theorem V found and repaired two errors in the discovering computation, and the result is reported here only in its corrected form. First, a kernel-form inconsistency had overstated the inhomogeneous-dilution cap, inflating the headline margins by ×2.5; the corrected irreducible gain is 0.94·S and the hostile-corner margin ×3.4, at which the kill survives. Second, the discovering run's *binary-star* horn — resonant anti-damping of field binaries with a claimed separation-gap sweep — is **nulled at ×2900 by spectral hole-burning** (in an inhomogeneously broadened medium the drain burns a Γ-slice hole rather than clamping the profile), and that observable is withdrawn; the repaired kill runs entirely through the gas continuum, for which hole-burning *is* the full-line clamp. A disk-heating corollary was checked and honestly does not kill ($\Delta v^2/v^2 \approx 4\times10^{-6}$ per 10 Gyr). The wide-binary $\gamma \longrightarrow 1.28$ prediction raised by the construction dies with the construction and is likewise withdrawn.

5. What this is, and is not

This chain does **not** prove "modified inertia is impossible": Theorem V closes the inverted-line realization — the unique sign-viable class the first four theorems permitted — and one passive-anharmonic corner remains parked on the prior sign-theorem adjudication. It does not touch the a_0 reframing ($a_0 = c^2\sqrt{(\Lambda/32\pi)}$ as a vacuum-set scale), the framework's live empirical tests (the fixed-direction $\bar{s}$^TX ephemeris fit; the $a_0(z)$ discriminants; the wide-binary four-way separation at Gaia DR4), or the modified-gravity realizations with their own known costs. What it establishes is narrower and, we believe, more useful: the covariant-modified-inertia programme now possesses a complete map — four theorems that force a unique surviving object, a constructive demonstration that the object passes every previously fatal gate, and a fifth theorem showing that the universe it must live in — a universe containing cold gas — destroys it. The forced-versus-posited ledger is explicit at every step, every script is committed and exits 0, and the corrections found in verification are printed beside the results they repair. If a sixth idea reopens this door, this chain says exactly which wall it must breach.

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